Question 1(a): Periodic and Nonperiodic Signals

[Unclear in screenshot: the original question sentence is not fully visible. Based on the visible answer table, the task is to determine whether each signal is periodic and, if periodic, find its fundamental period.]

i. $x(t) = \cos\left(t + \frac{\pi}{4}\right)$ (4 marks)

ii. $x(t) = \sin\left(\frac{2\pi}{3}t\right)$ (4 marks)

iii. $x[n] = e^{j(\pi/4)n}$ (4 marks)

iv. $x[n] = \cos\left(\frac{1}{4}n\right)$ (4 marks)

Question 1(c): Discrete-Time Signal Transformations

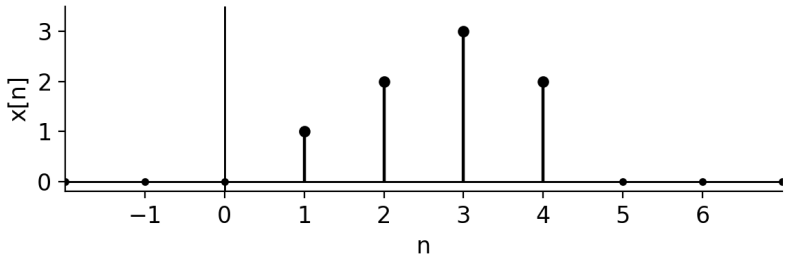
[Unclear in screenshot: the first half of the sentence is missing. Visible text says: "following transformations in (i) and (ii). Describe the changes in the signal for each case."]

Given the signal shown in Figure Q1.c, describe the following transformations:

i. $x[n-2]$ (6 marks)

ii. $x[-n]$ (6 marks)

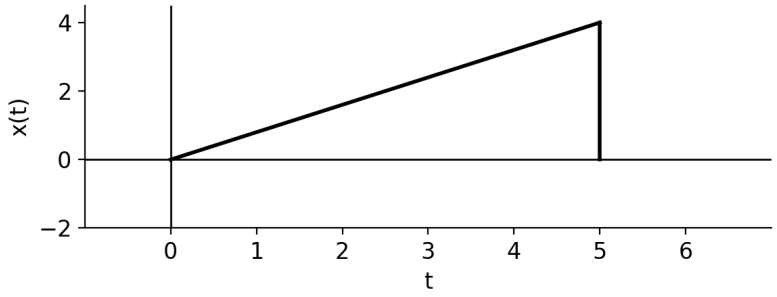
Figure Q1.c



Question 1(d): Even and Odd Components

Illustrate how the even and odd components of the signal shown in Figure Q1.d can be identified and described. Explain the characteristics of each component based on the original signal. (6 marks)

Figure Q1.d



Question 1(e): Energy and Power Signals

Explain how energy signals and power signals differ. Based on their characteristics, classify each of the following signals as an energy signal, a power signal, or neither.

i. $x(t) = e^{-at}u(t),\ a > 0$ (2 marks)

ii. $x(t) = tu(t)$ (2 marks)

Question 2(a): Convolution with a Unit Step Input

QUESTION 2 (50 Marks) **[CLO2-PLO2:C4]**

The input $x(t)$ and the impulse response $h(t)$ of a continuous-time LTI system are given by

$$x(t) = u(t), \qquad h(t) = e^{-\alpha t}u(t), \quad \alpha > 0.$$

i. Determine the output $y(t)$ by the equation

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau.$$

(12 marks)

ii. Determine the output $y(t)$ by the equation

$$y(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau.$$

(8 marks)

Question 2(b): Convolution of Two Rectangular Signals

Find the convolution of the two signals $X_1(t)$ and $X_2(t)$ as shown in Figure Q2.b. Your solution must include:

- Mathematical expressions for $X_1(t)$ and $X_2(t)$.
- Graphical illustration of the time-reversal and shifting of one signal. Show key stages of overlap.
- Derivation of the convolution integral with clearly defined integration limits for all relevant time intervals.
- Evaluation of the integrals to find the piecewise expression for $y(t)$.
- A clear sketch of the resulting convolved signal $y(t)$, labelling axes and key values.
- Show all your working steps clearly for a full mark.

(30 Marks)

